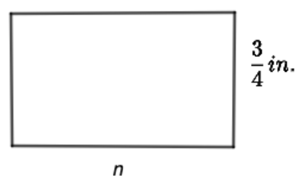
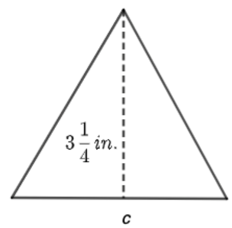
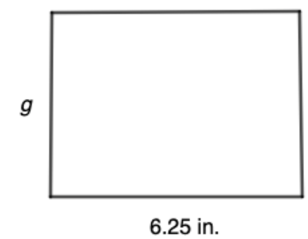
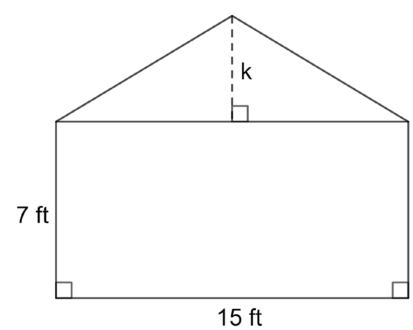
Grade 6
Unit 8
Lesson 5 Practice

Name______________________
Date______________________
Period____________________

Find the missing dimensions of the quadrilaterals and triangle. All measurements are in inches (in.).

 $4.5 \div 0.75 = 6$	 $13 \div 3.25 \div 0.5 = 8$	 $25 \div 6.25 = 4$
Area $= 4.5$ in^2	Area $= 13$ in^2	Area $= 25$ in^2
1. $n =$ 6 in	2. $c =$ 8 in	3. $g =$ 4 in

A composite figure is shown measured in feet (ft).



4. Determine the area of the triangle. Include units.
   Triangle $= 0.5 \cdot 15 \cdot k$
   Area$= 7.5k$ ft^2

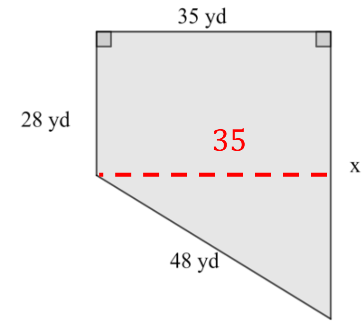
5. If the area of the composite figure is 150 ft^2, what is the area of the rectangle? Include units.
   Rectangle: $15 \cdot 7 = 105$ ft^2

6. What is the value of the missing dimension, k? Include units.

   $7.5k = 150 - 105$
   $7.5k = 45$
   $k = 6$ ft

7. A university is preparing for the upcoming spring sports season by preparing their grounds. The field is shown. The field has an area of $1{,}190$ square yards (yd^2). Determine the missing dimension, x. Include units.

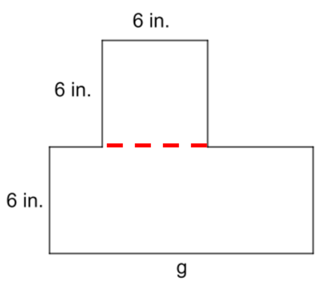
Rectangle: $28 \cdot 35 = 980$ yd^2
Triangle: $35 \cdot (x - 28) \cdot 0.5 = 17.5(x - 28)$ yd^2



$$\begin{array}{r} 1190 \\ \underline{-\ 980} \\ 210 \end{array}$$

$17.5(x - 28) = 210$
$x - 28 = 21$
$x = 40$ yd

8. A composite figure with an area of 108 square inches (in^2) is shown. Determine the missing dimension, g. Include units.



Square: $6 \cdot 6 = 36$ in^2

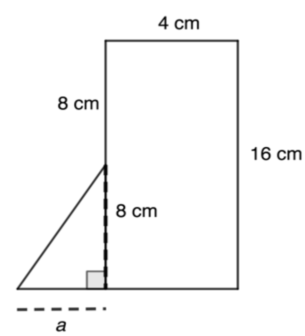
Rectangle: $6 \cdot g$ in^2

$$\begin{array}{r} 108 \\ \underline{-\ 36} \\ 72 \end{array}$$

$6g = 72$
$g = 12$ in.

9. A composite figure with an area of 76 square centimeters (cm^2). Determine the missing dimension, a. Include units.

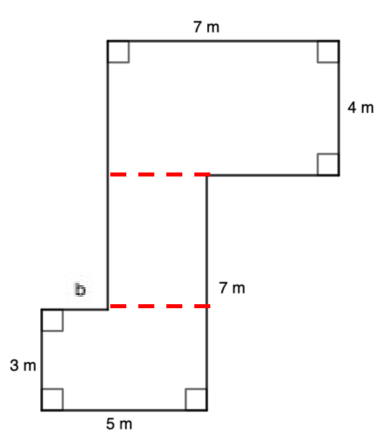
Rectangle: $16 \cdot 4 = 64$ cm^2
Triangle: $8 \cdot a \cdot 0.5 = 4a$ cm^2



$$\begin{array}{r} 76 \\ \underline{-64} \\ 12 \end{array}$$

$4a = 12$
$a = 3$ cm

10. Jackson is planning the foundation for a new office building. The foundation has an area of 51 square meters (m^2). Determine the missing dimension, b. Include units.



Top: $7 \cdot 4 = 28$ m^2
Middle: $4(b - 5)$ m^2
Bottom: $3 \cdot 5 = 15$ m^2
$28 + 15 = 43$ m^2

$$\begin{array}{r} 51 \\ \underline{-43} \\ 8 \end{array}$$

$4(5 - b) = 8$
$20 - 4b = 8$
$b = 3$